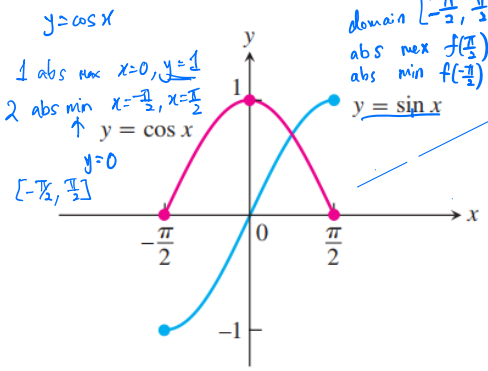
Application of Derivatives

Extreme Values of Functions

Definitions. *Absolute Maximum, Absolute Minimum*
Let f be a function with domain D. Then f has **absolute maximum** value on D at a point c if
$$f(x) \le f(c) \quad \text{for all } x \text{ in } D$$
and an **absolute minimum** value on D at c if
$$f(x) \ge f(c) \quad \text{for all } x \text{ in } D.$$

Maximum and minimum values are called **extreme values** or **extrema** (plural of the Latin *extremum*). Absolute maxima or minima (absolute extrema) are also called **global** maxima or minima **(global** extrema), to distinguish them from *local extrema*.

Example.

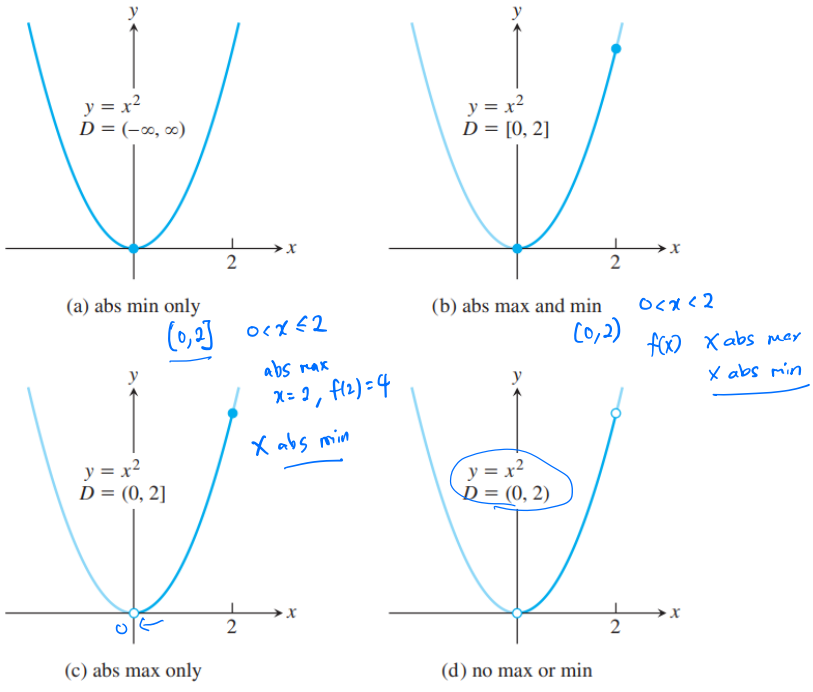


On closed interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the function $f(x) = \cos x$ takes on an absolute maximum value of 1 (once) and an absolute minimum value of 0 (twice).

On the same interval, the function $g(x) = \sin x$ takes on a maximum value of 1 and a minimum value of -1.

Functions with the same defining rule can have different extrema, depending on the domain.

Example.



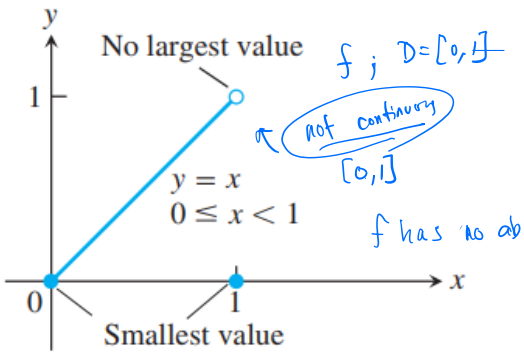
Function rule	Domain D	Absolute Extrema on D
(a) $y = x^2$	$(-\infty, \infty)$	No absolute maximum. Absolute minimum of 0 at $x = 0$.
(b) $y = x^2$	$[0,2]$	Absolute maximum of 4 at $x = 2$. Absolute minimum of 0 at $x = 0$.
(c) $y = x^2$	$(0,2]$	Absolute maximum of 4 at $x = 2$. No absolute minimum.
(d) $y = x^2$	$(0,2)$	No absolute extrema.

Theorem 1. *The Extreme Value Theorem*

If f is *continuous* on a closed interval $[a, b]$, then f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$.
That is, there are numbers x_1 and x_2 in $[a, b]$ with $f(x_1) = m, f(x_2) = M$, and $m \le f(x) \le M$ for every other x in $[a, b]$.

Note: The requirements in the Theorem that the interval be closed and finite, and that the function be continuous, are *key ingredients*. Without them, the conclusion of the theorem need not hold.

Example.



Even a *single point of discontinuity* can keep a function from having either a maximum or minimum value on a closed interval.

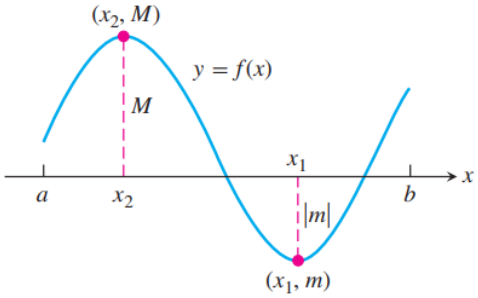
The function

$$y = \begin{cases} x, & 0 \le x < 1 \\ 0, & x = 1 \end{cases}$$

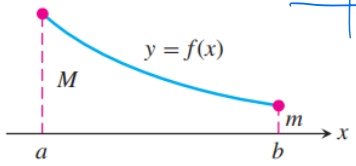
is continuous at every point of $[0,1]$ except $x = 1$, yet its graph over $[0,1]$ *does not have a highest point*.

Note: The example above shows that the continuity requirement cannot be omitted.

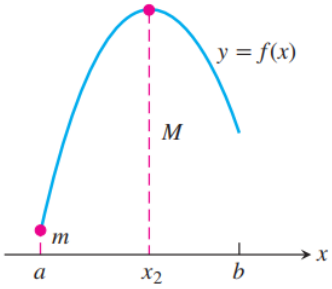
Example. The following is some possibilities for a continuous function's maximum and minimum on a closed interval $[a, b]$.



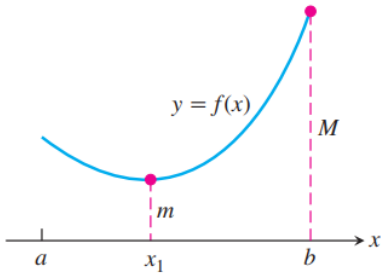
Maximum and minimum at interior points



Maximum and minimum at endpoints



Maximum at interior point, minimum at endpoint



Minimum at interior point, maximum at endpoint

Local (Relative) Extreme Values

Definitions *Local Maximum, Local Minimum*

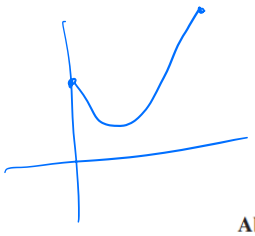
A function f has a **local maximum** value at an interior point c of its domain if

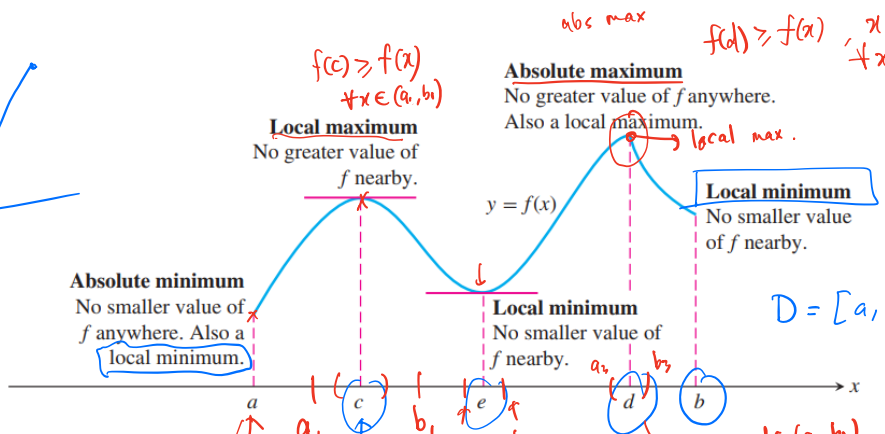
$f(x) \le f(c)$ for all x in some open interval containing c.

A function f has a **local minimum** value at an interior point c of its domain if

$f(x) \ge f(c)$ for all x in some open interval containing c.

Example.





The function's *absolute minimum* occurs at a even though at e the function's value is smaller than at any other point *nearby*. The curve rises to the left and falls to the right around c, making $f(c)$ a maximum locally. The function attains its absolute maximum at d.
The function f has *local maxima* at c and d and *local minima* at a, e, and b. Local extrema are also called **relative extrema**.

Finding Extrema.

The next theorem explains why we usually need to investigate only a few values to find a function's extrema.

Theorem 2. *The First derivative Theorem for Local Extreme Values.*
If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c, then

$$f'(c) = 0.$$

The Theorem says that a *function's first derivative is always zero* at an interior point where the function has a local extreme value and the derivative is defined.

Hence the only places where a function f can possibly have an extreme value (local or global) are

1. interior points where $f' = 0$,
2. interior points where f' is undefined,
3. endpoints of the domain of f.

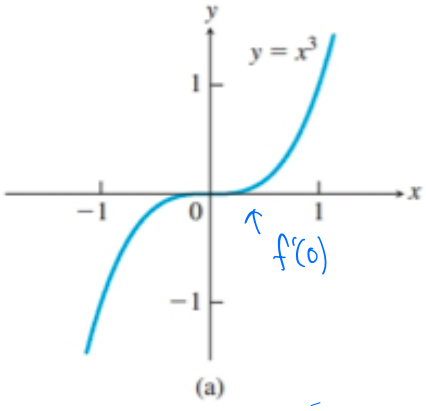
The following definition helps us to summarize.

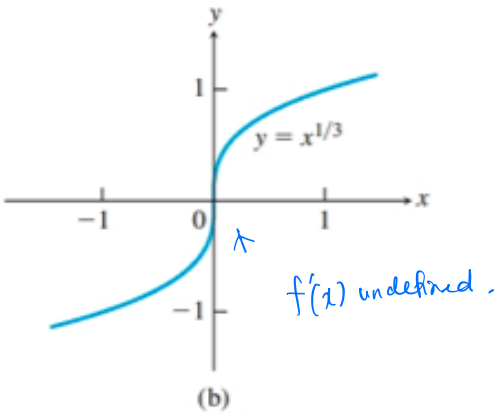
Definition. *Critical Point*
An interior point of the domain of a function f where f' *is zero or undefined* is a **critical point** of f.

Note: the only domain points where a function can assume extreme values are *critical points* and ~~*endpoints*~~.

We note that, the converse of the **Theorem** is not necessarily true. That is, a differentiable function may have a critical point at $x = c$ without having a local extreme value there.

Example. The function $f(x) = x^3$ has a critical point at $x = 0$, where $f(0) = 0$, but is positive to the right of the origin and negative to the left. So, it cannot have a local extreme value at the origin. Instead, it has a *point of inflection* there. Similarly, $y = x^{\frac{1}{3}}$.





If the interval is not closed or not finite (such as $a < x < b$ or $a < x < \infty$), we have seen that absolute extrema need not exist.

However, many problems that ask for extreme values call for finding the absolute extrema of a continuous function on a closed and finite interval.

Theorem 1 assures us that such values exist; **Theorem 2** tells us that they are taken on only at critical points and endpoints.

How to Find the Absolute Extrema of a Continuous Function f on a Finite Closed Interval.

1. Find all critical points of f on the interval.
2. Evaluate f at all critical points and endpoints.
3. Take the largest and smallest of these values.

abs max ↑ abs min

Example. *Finding Absolute Extrema*
Find the absolute maximum and minimum values of $f(x) = x^2$ on $[-2,1]$.

Solution.

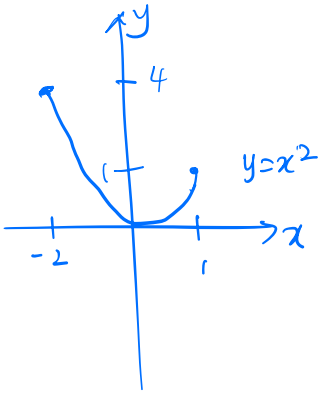
$f(x) = x^2$, $[-2, 1]$

$f'(x) = 2x = 0 \Rightarrow x = 0$ critical point

Critical point: $f(0) = 0$ (abs. min).

endpoints: $x = -2$, $f(-2) = 4$ → abs max

$x = 1$, $f(1) = 1$



Example. *Absolute Extrema at Endpoints*

Find the absolute extrema values of $g(t) = 8t - t^4$ on $[-2,1]$.

Solution.

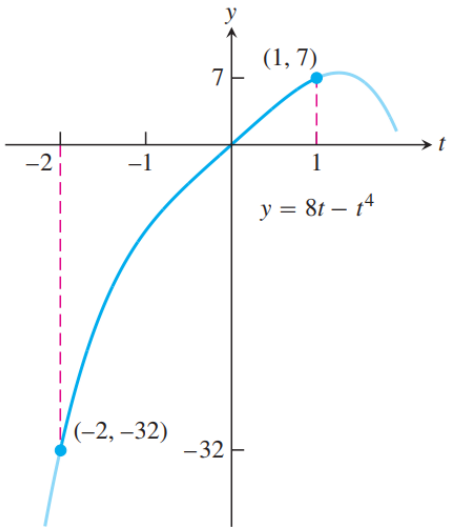
$g(t) = 8t - t^4$

$g'(t) = 8 - 4t^3$

$g'(t) = 0 \Rightarrow 4t^3 = 8$

$\Rightarrow t^3 = 2$

$= t = \sqrt[3]{2} > 1$ (not in domain)



$D = [-2, 1]$

$g(-2) = -16 - 16 = -32$ (abs min)

$g(1) = 8 - 1 = 7$ (abs max).

Example. *Finding Absolute Extrema on a Closed Interval.*

Find the absolute maximum and minimum values of $f(x) = x^{\frac{2}{3}}$ on the interval $[-2,3]$.

Solution.

$f(x) = x^{2/3}$

$f'(x) = \frac{2}{3x^{1/3}}$ $\quad f'(x) \neq 0$.

$f'(x)$ doesn't exist $\Rightarrow x = 0$.

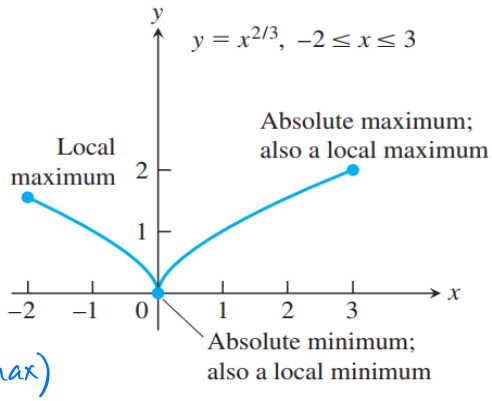
critical point : $x = 0$

$f(0) = 0$ (abs min).

endpoints : $x = -2$, $x = 3$

$f(-2) = (-2)^{2/3} = \sqrt[3]{4} = 1.5874$

$f(3) = 3^{2/3} = \sqrt[3]{9} = 2.0801$ (abs max)

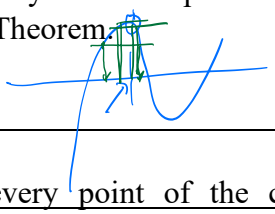


The Mean Value Theorem

We know that constant functions have zero derivatives, but could there be a complicated function, with many terms, the derivatives of which all cancel to give zero?

What is the relationship between two functions that have identical derivatives over an interval? What we are really asking here is what functions can have a particular *kind* of derivative.

These and many other questions we study in this chapter are answered by applying the Mean Value Theorem. We first need Rolle's Theorem.



f continuous [a,b]
f differentiable (a,b)
f(a) = f(b)
⇒ ∃ c ∈ (a,b) s.t. f'(c) = 0

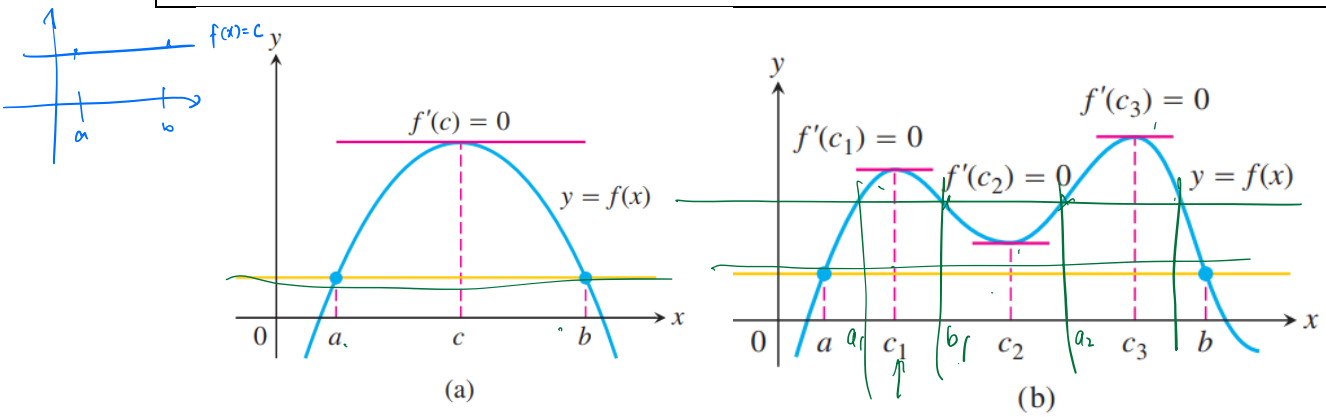
Rolle's Theorem.

Theorem. *Rolle's Theorem*
Suppose that $y = f(x)$ is continuous at every point of the closed interval $[a, b]$ and differentiable at every point of its interior (a, b). If

$$f(a) = f(b),$$

then there is at least one number c in (a, b) at which

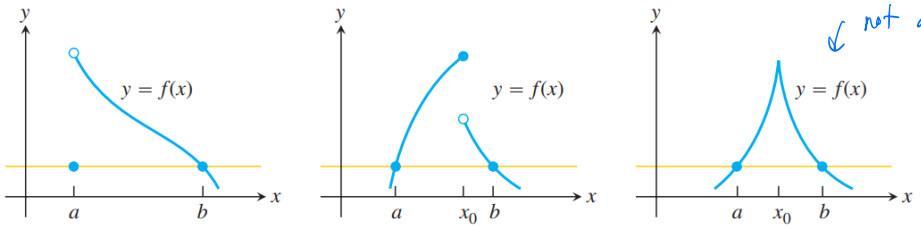
$$f'(c) = 0.$$



Rolle's Theorem says that a differentiable curve has at least one horizontal tangent between any two points where it crosses a horizontal line. It may have just one as in (a), or it may have more than one as in (b).

Note: The hypotheses of the *Rolle's Theorem* are essential. If they fail at even one point, the graph may not have a horizontal tangent.

Example. *The Hypotheses of Rolle's Theorem Do not Hold.*
There may be *no horizontal tangents* if the hypotheses of Rolle's Theorem do not hold.



f not cont [a,b]
✗ Rolle's thm

f not continuous [a,b]

not differentiable (a,b)

$f(x) = \frac{x^2-1}{x+1}$

f not cont. at $x=-1$

f not diff at $x=-1$

$f(x) = \frac{x^3}{3} - 3x$

f is a polynomial

$\Rightarrow f$ continuous, f differentiable $\mathbb{R}$.

$f(-3) = \frac{-3^3}{3} - 3(-3) = 0$

$f(3) = \frac{3^3}{3} - 3(3) = 0$

$\Rightarrow f(-3) = f(3)$, $[-3,3]$

$\exists\, c \in (-3,3)$ s.t. $f'(c) = 0 \Rightarrow$

$f'(x) = \frac{3x^2}{3} - 3$

$= x^2 - 3$

$f'(x) = 0$

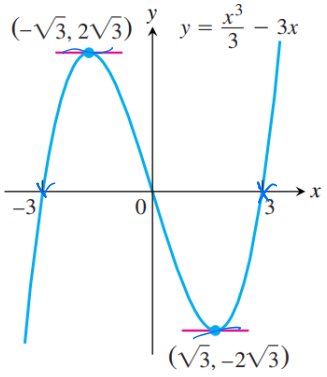
$\Rightarrow x = \pm\sqrt{3}$

$x = \pm 1.732 \in (-3,3)$

Example. *Horizontal Tangents of a Cubic Polynomial*
The polynomial function

$$f(x) = \frac{x^3}{3} - 3x$$

is continuous at every point of $[-3,3]$ and is differentiable at every point of $(-3,3)$. Since $f(-3) = f(3) = 0$. According to Rolle's Theorem, f' must be zero at least once in the open interval between $a = -3$ and $b = 3$.
In fact, $f'(x) = x^2 - 3$ is zero twice in this interval (see the figure below), once at $x = -\sqrt{3}$ and again at $x = \sqrt{3}$.



Rolle's Theorem may be combined with the Intermediate Value Theorem to show when there is only one real solution of an equation $f(x) = 0$.

Example. *Solution of an Equation* $f(x) = 0$.
Show that the equation $x^3 + 3x + 1 = 0$ has exactly one real solution.

Solution.

$f(x) = x^3 + 3x + 1$

f is a poly deg. of 3

$\Rightarrow f$ continuous and differentiable on $\mathbb{R}$.

$f(0) = 0^3 + 3(0) + 1 = 1 > 0$

$f(-1) = -1 + 3(-1) + 1 = -3 < 0$

By Intermediate Value Theorem, f has at least one roots in $\mathbb{R}$.

Assume $f(a) = f(b) = 0$ $a, b \in \mathbb{R}$.

By Rolle's Thm, $\exists\, c \in (a,b)$ s.t.

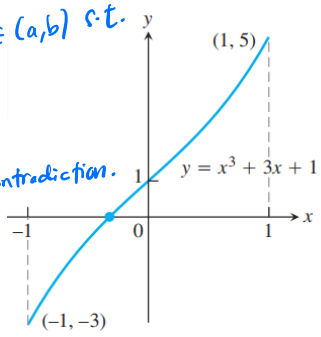
$f'(c) = 0$

$f'(x) = 3x^2 + 3 \geq 3$

$f'(x)$ never be zero; a contradiction.

$\Rightarrow f$ has only one real root.

$\Rightarrow$ the equation has only one real solution.



Extra exercises:

1. Prove that $f(x) = x^7 + x^5 + 2x - 1$ has only one real root.

2. Show that the equation $3\tan x + x^3 = 2$ has exactly one solution in the interval $\left[0, \frac{\pi}{4}\right]$.

$f(x) = 3\tan x + x^3 - 2 = 0$, $[0, \frac{\pi}{4}]$

f continuous and differentiable $[0, \frac{\pi}{4}]$

$f(0) = 3\tan 0 + 0^3 - 2 = -2 < 0$

$f(\frac{\pi}{4}) = 3\tan\frac{\pi}{4} + (\frac{\pi}{4})^3 - 2 = 1 + (\frac{\pi}{4})^3 > 0$

$\Rightarrow$ By Intermediate Value Theorem, f has at least 1 zero in $(0, \frac{\pi}{4})$

Assume $f(a) = f(b) = 0$, $[a,b] \subseteq (0, \frac{\pi}{4})$

By Rolle's theorem, $\exists\, c \in (a,b)$ s.t.

$f'(c) = 0$

$f'(x) = 3\sec^2 x + 3x^2$

$= 3(1 + \tan^2 x) + 3x^2$

$= 3 + 3\tan^2 x + 3x^2 \geq 3$ $[0, \frac{\pi}{4}]$

$f'(x)$ can never be zero $\Rightarrow$ a contradiction

$\Rightarrow$ f has exactly 1 zero in $[0, \frac{\pi}{4}]$.

$\Rightarrow$ The eqn has only 1 solution in $[0, \frac{\pi}{4}]$.



Rolle's Thm

if f continuous $[a,b]$,

f differentiable (a,b),

$f(a) = f(b)$,

then $\exists\, c \in (a,b)$ s.t.

$f'(c) = 0$.

Mean Value Theorem,

if f continuous $[a,b]$,

f differentiable (a,b), $\Rightarrow f(a) = f(b)$

then $\exists\, c \in (a,b)$ s.t.

$f'(c) = \frac{f(b) - f(a)}{b-a}$.

$f(a) = f(b)$

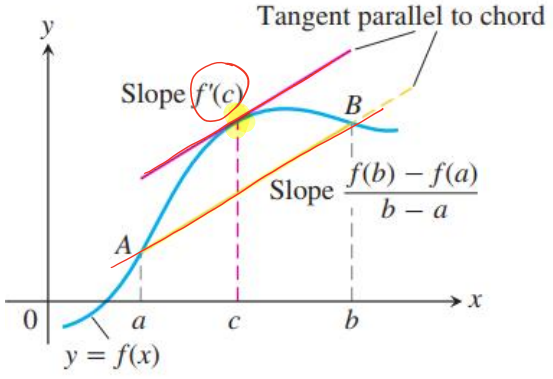
$f'(c) = \frac{f(b) - f(a)}{b-a} = \frac{0}{b-a} = 0$

The Mean Value Theorem

Theorem. *The Mean Value Theorem*

Suppose $y = f(x)$ is continuous on a closed interval $[a, b]$ and differentiable on the interval's interior (a, b). Then there is at least one point c in (a, b) at which

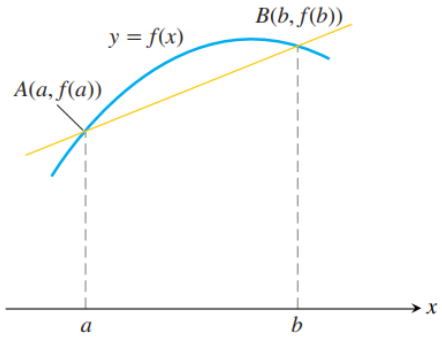
$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Geometrically, the Mean Value Theorem says that somewhere between A and B the curve has at least one tangent parallel to chord AB.

$\frac{dy}{dx} = ?$

Proof. We picture the graph of f as a curve in the plane and draw a line through the points $A\,(a, f(a))$ and $B(b, f(b))$. (Figure below)



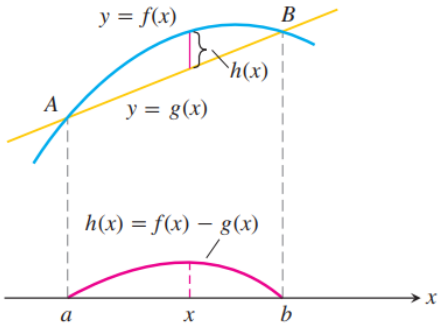
The graph of f and the chord AB over the interval $[a, b]$.

The line is the graph of the function (line equation)

$$g(x) = f(a) + \frac{f(b) - f(a)}{b - a}\,(x - a)$$

The vertical difference between the graphs of f and g at x is

$$\begin{aligned} h(x) &= f(x) - g(x) \\ &= f(x) - f(a) - \frac{f(b) - f(a)}{b - a}\,(x - a) \end{aligned}$$



The chord AB is the graph of the function $g(x)$.

The function $h(x) = f(x) - g(x)$ gives the vertical distance between the graphs of f and g at x.

The function h satisfies the hypotheses of Rolle's Theorem on $[a, b]$. It is continuous on $[a, b]$ and differentiable on (a, b), because both f and g are. Also $h(a) = h(b) = 0$, it is because the graphs of f and g both pass through A and B.
Therefore, $h'(c) = 0$ at some point $c \in (a, b)$. Hence,

$$h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

$$h'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}$$

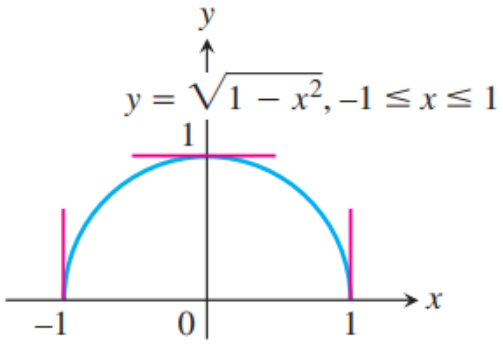
$$0 = f'(c) - \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

This completes the proof. ■

Note. The hypotheses of the Mean Value Theorem do not require f to be differentiable at either a and b. Continuity at a and b is enough. (See example in the following.)

Example.



The function $f(x) = \sqrt{1-x^2}$ satisfies the hypotheses (and conclusion) of the Mean Value Theorem on $[-1,1]$ even though f is not differentiable at -1 and 1.

Example.
The function $f(x) = x^2$ is continuous for $0 \le x \le 2$ and differentiable for $0 < x < 2$. Since $f(0) = 0$ and $f(2) = 4$, the *Mean Value Theorem* says that at some point c in the interval, the derivative $f'(x) = 2x$ must have the value $\frac{4-0}{2-0} = 2$. In this (exceptional) case we can identify c by solving the equation $2c = 2$ to get $c = 1$.

A Physical Interpretation

We can think of the number $\frac{f(b)-f(a)}{b-a}$ as the average change in f in f over $[a, b]$ and can view $f'(c)$ as an instantaneous change.
Then the Mean Value Theorem says that the *instantaneous change at some interior point is equal to the average change over the entire interval*.

Example. If a car accelerating from zero takes 8s to go 176m, its average velocity for the 8s interval is $\frac{176}{8} = 22\ m/s$.
The Mean Value Theorem says that at some point during the acceleration the speedometer must read exactly 79.2km/h (or 22m/s).

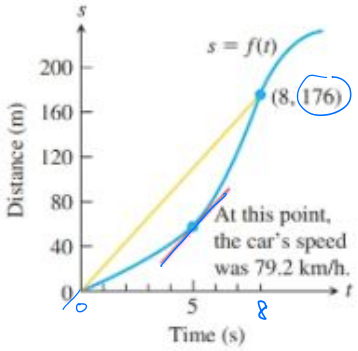


FIGURE 4.18 Distance versus elapsed time for the car in Example 3.

Mathematical Consequences

What kind of function has a zero derivative over an interval?
The first corollary of the Mean Value Theorem provides the answer that only constant functions have zero derivatives.

Corollary.
If $f'(x) = 0$ at each point x in open interval (a, b), then $f(x) = C$ for all $x \in (a, b)$, C is a constant.

Proof: Show f has a constant value on the interval (a, b). Now f satisfies the hypothesis of Mean Value Theorem on $[x_1, x_2]$. It is differentiable at every point of $[x_1, x_2]$ and hence continuous at every point as well. Therefore,

$$f'(c) = \frac{(f(x_1) - f(x_2))}{x_2 - x_1}$$

At some point $c \in (x_1, x_2)$. Since $f' = 0$ throughout (a, b), this equation implies successively that

$$\frac{f(x_1) - f(x_2)}{x_2 - x_1} = 0 \Rightarrow f(x_1) = f(x_2).$$

Corollary. *Functions with the Same Derivative Differ by a Constant*
If $f'(x) = g'(x)$ at each point x in open interval (a, b), then there exists a constant C such that $f(x) = g(x) + C$ for all $x \in (a, b)$. That is, $f - g$ is a constant on (a, b).

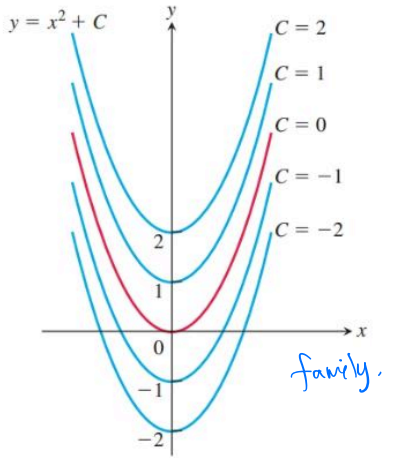


FIGURE 4.19 From a geometric point of view, Corollary 2 of the Mean Value Theorem says that the graphs of functions with identical derivatives on an interval can differ only by a vertical shift. The graphs of the functions with derivative $2x$ are the parabolas $y = x^2 + C$, shown here for several values of C.

Proof. At each point $x \in (a, b)$ the derivative of $h = f - g$ is

$$h'(x) = f'(x) - g'(x) = 0$$

Thus, $h(x) = C$ on (a, b). That is, $f(x) = g(x) + C$ on (a, b).

derivative is $\sin x$

$g(x) = -\cos x$, $g'(x) = \sin x$

$f'(x) = g'(x) \Rightarrow f(x) = -\cos x + C$

$x = 0$, $f(0) = 2 \Rightarrow 2 = -\cos 0 + C$

$= C - 1$

$C = 3$

$f(x) = 3 - \cos x$.

Example.
Find the function $f(x)$ whose derivative is $\sin x$ and whose graph passes through the point $(0,2)$.

Solution. Since $f(x)$ has the same derivative as $g(x) = -\cos x$, we know that $f(x) = -\cos x + C$ for some constant C. The value of C can be determined from the condition that $f(0) = 2$ (the graph of f passes through $(0,2)$):

$$f(0) = -\cos 0 + C = 2.$$

So, $C = 3$. The function is $f(x) = -\cos x + 3$.

Example:

1. Show that for any numbers a and b, the sine inequality $|\sin b - \sin a| \le |b - a|$ is true.
2. Show that $\tan x > x$ for all $x \in \left(0, \frac{\pi}{2}\right)$.

$\Rightarrow \left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, $b \neq 0$

① Show: $|\sin a - \sin b| \le |b - a|$, $a, b \in \mathbb{R}$

$f(x) = \sin x$, $[a,b]$

f continuous $[a,b]$,

f differentiable (a,b)

By Mean Value Theorem,

$\exists c \in (a,b)$ s.t.

$f'(c) = \frac{\sin b - \sin a}{b - a}$

$f'(x) = \cos x$

$f'(c) = \cos c$, $c \in (a,b)$

$-1 \le \cos c \le 1$

$\Rightarrow |\cos c| \le 1$

$\Rightarrow \left|\frac{\sin b - \sin a}{b - a}\right| \le 1$

$\Rightarrow |\sin b - \sin a| \le |b - a|$.

$y = x^x$

$y = x^{x^x}$

$y = x^{x^{x^{\cdots}}}$

② $\tan x > x$, $\forall x \in (0, \frac{\pi}{2})$.

$f(\theta) = \tan\theta$ $[0,x]$, $x \in (0, \frac{\pi}{2})$.

f continuous $[0,x]$

f differentiable $(0,x)$.

By Mean Value Theorem,

$\exists c \in (0,x)$ s.t.

$f'(c) = \frac{\tan x - \tan 0}{x - 0}$

$= \frac{\tan x}{x}$

$\theta \in (0, \frac{\pi}{3})$

$f'(\theta) = \sec^2\theta = 1 + \tan^2\theta$

$f'(\theta) > 1$

$\Rightarrow \frac{\tan x}{x} > 1$, $x \in (0, \frac{\pi}{2})$

$\Rightarrow \tan x > x$ //

Monotonic Functions and The First Derivative Test

In sketching the graph of a differentiable function, it is useful to know where it increases (rises from left to right) and where it decreases (falls from left to right) over an interval.

This section defines precisely what it means for a function to be increasing or decreasing over an interval and gives a test to determine where it increases and where it decreases. We also show how to test the critical points of a function for the presence of local extreme values.

Increasing Functions and Decreasing Functions

Another corollary to the Mean Value Theorem.

A function that is either increasing on an interval or decreasing on an interval is said to be **monotonic** on the interval.

f increasing $[0,4] \rightarrow f$ monotonic $[0,4]$

f decreasing $[4,8] \rightarrow f$ monotonic $[4,8]$

f not monotonic $[0,8]$

Corollary. *Increasing, Decreasing Function*

Suppose that f is continuous on $[a, b]$ and differentiable on (a, b).
1. If $f'(x) > 0$ for each point $x \in (a, b)$, then f is **increasing** on $[a, b]$.
2. If $f'(x) < 0$ for each point $x \in (a, b)$, then f is **decreasing** on $[a, b]$.

A function that is increasing or decreasing on I is called **monotonic** on I.

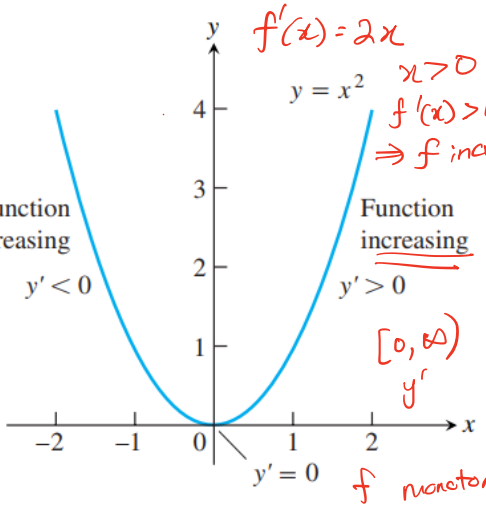
Proof. Let x_1 and x_2 be any two points in $[a, b]$ with $x_1 < x_2$. The Mean Value Theorem applied to f on $[x_1, x_2]$ says that

$$f(x_1) - f(x_2) = f'(c)(x_2 - x_1)$$

For some c between x_1 and x_2. The sign of the right-hand side of this equation is the same as the sign of $f'(c)$ because $x_2 - x_1$ is positive.

Corollary tells us that $f(x) = \sqrt{x}$ is increasing on the interval $[a, b]$ for any $b > 0$ because $f'(x) = \frac{1}{2\sqrt{x}}$ is positive on $(0, b)$. The derivative does not exist at $x = 0$, but Corollary still applies. The corollary is also valid for open and for infinite intervals, so $f(x) = \sqrt{x}$ is also increasing on $(0,1)$, on $(0, \infty)$ and on $[0, \infty)$.

Example. The function $f(x) = x^2$.



From its graph:
Decreases on $(-\infty, 0]$
Increases on $[0, \infty)$
Monotonic on $(-\infty, 0]$ and $[0, \infty)$,
But not monotonic on $(-\infty, \infty)$.

Note: on $(-\infty, 0)$ the tangents have *negative* slopes, so the first derivative is always negative. For $(0, \infty)$ the tangents have positive slopes and first derivative is *positive*.

To find the intervals where a function f is increasing or decreasing, we first find all of the critical points of f. If $a < b$ are two critical points for f, and if the derivative f' is continuous but never zero on the interval (a, b), then by Intermediate Value Theorem applied to f', the derivative must be everywhere positive on (a, b), or everywhere negative there.

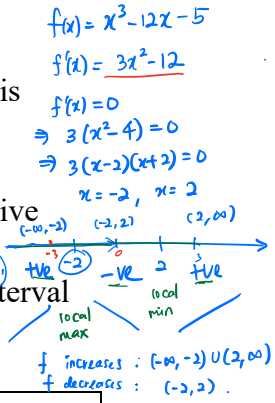
One way we can determine the sign of f' on (a, b) is simply by evaluating the derivative at a single point c in (a, b).

If $f'(c) > 0$, then $f'(x) > 0$ for all x in (a, b). So f is increasing on$[a, b]$;
If $f'(c) < 0$, then $f'(x) > 0$ for all x in (a, b). So f is decreasing on$[a, b]$.

It doesn't matter which point c we choose in (a, b), since the sign of $f'(c)$ is the same for all choices. Usually we pick c to be a point where it is easy to evaluate $f'(c)$.

Corollary. *First Derivative Test for Monotonic Functions.*
Suppose that f is continuous on $[a, b]$ and differentiable on (a, b).
If $f'(x) > 0$ at each point $x \in (a, b)$, then f is increasing on $[a, b]$.
If $f'(x) < 0$ at each point $x \in (a, b)$, then f is decreasing on $[a, b]$.

Example. *Using the First Derivative Test for Monotonic Functions*
Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the intervals on which f is increasing and decreasing.

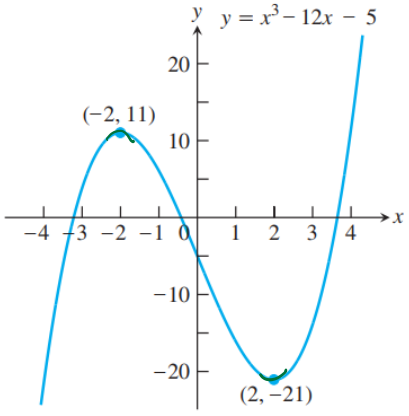


Solution. The function f is everywhere continuous and differentiable. The first derivative

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2).$$

f is zero at $x = -2$ and $x = 2$. These critical points subdivide the domain of f into interval $(-\infty, -2)$, $(-2,2)$, and $(2, \infty)$ on which f' is either positive or negative.

Intervals	$-\infty < x < 2$	$-2 < x < 2$	$2 < x < \infty$
f' Evaluated	$f'(-3) = 15$	$f'(0) = -12$	$f'(3) = 15$
Sign of f'	+	-	+
Behaviour of f	Increasing	Decreasing	Increasing



Note: The function $f(x) = x^3 - 12x - 5$ is monotonic on three separate intervals.

First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f, and that f is differentiable at every point in some interval containing c except possibly at c itself.
Moving across c from left to right,

1. if f' changes from negative to positive at c, then f has a *local minimum* at c;
2. if f' changes from positive to negative at c, then f has a *local maximum* at c;
3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c.

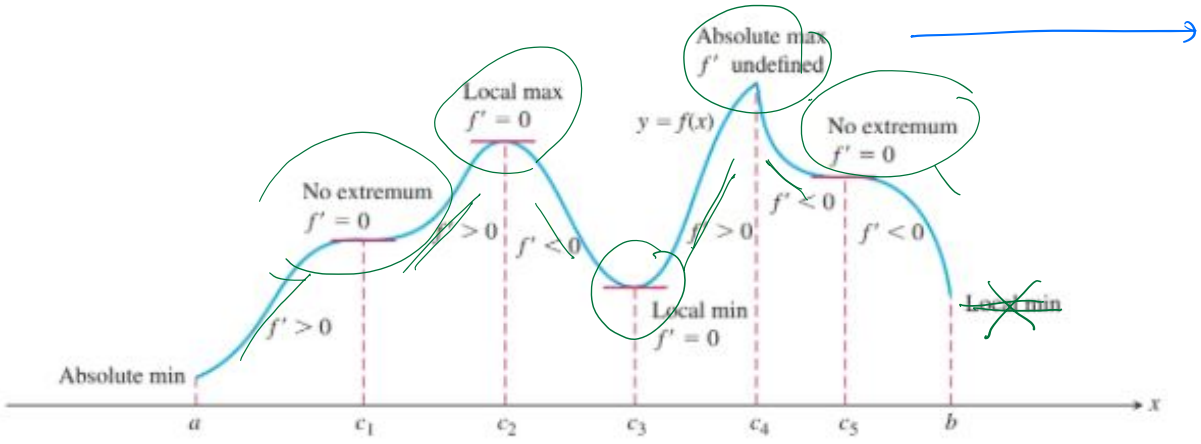


FIGURE 4.21 The critical points of a function locate where it is increasing and where it is decreasing. The first derivative changes sign at a critical point where a local extremum occurs.

Example. *Using the First Derivative Test for Local Extrema*

Find the critical points of

$$f(x) = x^{\frac{1}{3}}(x-4) = x^{\frac{4}{3}} - 4x^{\frac{1}{3}}.$$

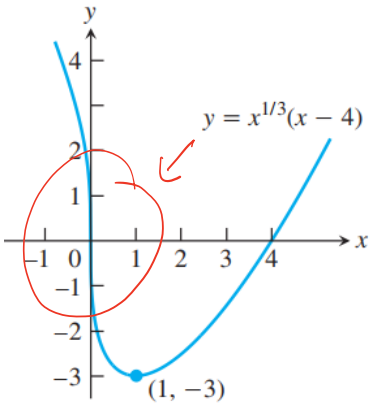
Identify the intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.

Solution. The function f is continuous at all x since it is the product of two continuous functions, $x^{\frac{1}{3}}$ and $(x-4)$. The first derivative

$$f'(x) = \frac{4}{3}x^{\frac{1}{3}} - \frac{4}{3}x^{-\frac{2}{3}} = \frac{4}{3}x^{-\frac{2}{3}}(x-1) = \frac{4(x-1)}{3x^{\frac{2}{3}}}.$$

Hence, $f'(x) = 0 \Rightarrow x = 1$ and undefined at $x = 0$. We note that it has no endpoints in the domain, so the critical points $x = 0$ and $x = 1$ are the online places where f might have an extreme value.

Intervals	$x < 0$	$0 < x < 1$	$x > 1$
Sign of f'	-	-	+
Behaviour of f	Decreasing	Decreasing	Increasing



The function f decreases on $(-\infty, 0)$, decreases on $(0,1)$, and increases on $(1, \infty)$.
According to the First Derivative Test for Local Extrema tells us that f does not have an extreme value at $x = 0$ (f' does not change sign) and that f has a local minimum at $x = 1$ (f' changes sign from negative to positive).

The value of local minimum of f is $f(1) = 1^{\frac{2}{3}}(1 - 4) = -3$.
It is also an absolute minimum.

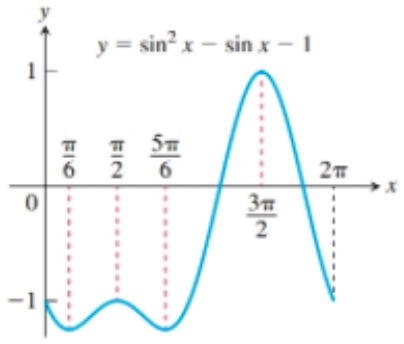
Example

Within the interval $0 \leq x \leq 2\pi$, find the critical points of

$$f(x) = \sin^2 x - \sin x - 1.$$

$[0, 2\pi]$

Identify the open intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.



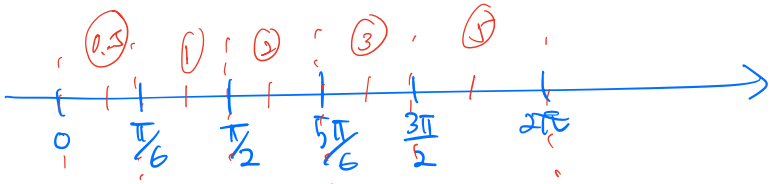
$f(x) = \sin^2 x - \sin x - 1$

$f'(x) = 2\sin x \cos x - \cos x = 0$

$\Rightarrow \cos x (2\sin x - 1) = 0$

$\Rightarrow \cos x = 0$, $\sin x = 1/2$. $x \in [0, 2\pi]$

$\Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$ $x = \frac{\pi}{6}, \frac{5\pi}{6}$.



Sign of $f'(x)$: -ve, +ve, -ve, +ve, -ve

f increases : $(\frac{\pi}{6}, \frac{\pi}{2}) \cup (\frac{5\pi}{6}, \frac{3\pi}{2})$

f decreases : $(0, \frac{\pi}{6}) \cup (\frac{\pi}{2}, \frac{5\pi}{6}) \cup (\frac{3\pi}{2}, 2\pi)$

local min : $(\frac{\pi}{6}, -\frac{5}{4})$, $(\frac{5\pi}{6}, -\frac{5}{4})$ = abs. min .

local max : $(\frac{\pi}{2}, -1)$, $(\frac{3\pi}{2}, 1)$. " abs max .

endpoints : $(0, -1)$

$(2\pi, -1)$

Concavity and Curve Sketching

In this section, we see how the second derivative gives information about the way the graph of a differentiable function bends or turns. This additional information enables us to capture key aspects of the behaviour of a function and its graph, and then present these features in a sketch of the graph.

Concavity

Definition. *Concave Up, Concave Down.*
The graph of a differentiable function $y = f(x)$ is
(a) **Concave up** on an open interval I if f' is *increasing* on I.
(b) **Concave down** on an open interval I if f' is *decreasing* on I.

A function whose graph is concave up is also often called **convex**.

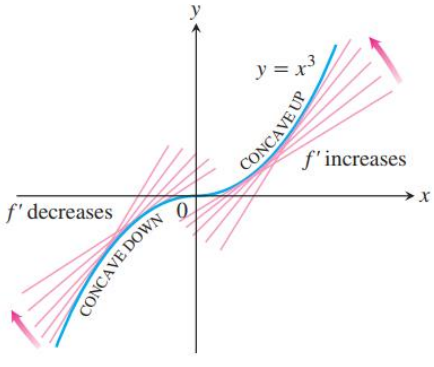
If $y = f(x)$ has a second derivative, we can apply the **Corollary** "*First Derivative Test for Monotonic Functions.*" To conclude that f' increases if $f'' > 0$ on I and decreases if $f'' < 0$.

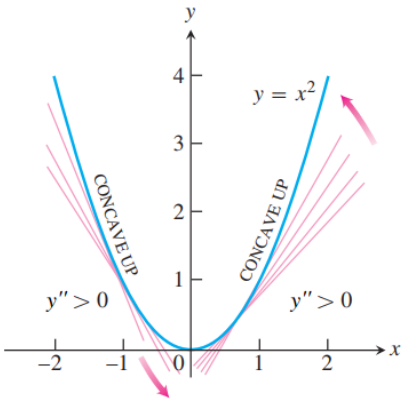
The Second Derivative Test for Concavity
Let $y = f(x)$ be twice-differentiable on an interval I.
1. If $f'' > 0$ on I, the graph of f over I is *concave up*.
2. If $f'' < 0$ on I, the graph of f over I is *concave down*.

Example. *Applying the Concave Test*

(a) The curve $y = x^3$ is concave down on $(-\infty, 0)$ where $y'' = 6x < 0$ and concave up on $(0, \infty)$ where $y'' = 6x > 0$.

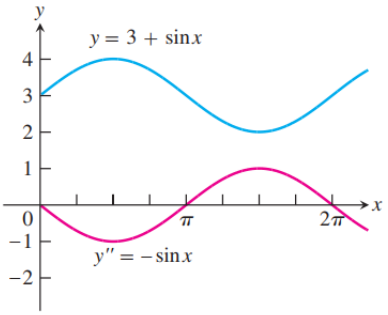
(b) The curve $y = x^2$ is concave up on $(-\infty, \infty)$ because its second derivative $y'' = 2 > 0$ is always positive.





Example. *Determining Concavity Test*
Determine the concavity of $y = 3 + \sin x$ on $[0,2\pi]$.

Solution. The graph of $y = 3 + \sin x$ is concave down on $(0, \pi)$, where $y'' = -\sin x$ is negative. It is concave up on $(\pi, 2\pi)$, where $y'' = -\sin x$ is positive.



Point of Inflection.

By referring the curve of $y = 3 + \sin x$, it changes concavity at the point $(\pi, 3)$. We call the point $(\pi, 3)$ as a *point of inflection* of the curve.

Definition. *Point of Inflection*
A point $(c, f(c))$ where the graph of a function has a tangent line and where the concavity changes is a **point of inflection**.

Note. A point on a curve where y'' is positive one side and negative on the other side is a point of inflection. At such point, y'' is either zero or undefined.
If y is a twice-differentiable function, $y'' = 0$ at a point of inflection and $y' = 0$ has a local maximum or minimum.

At a point of inflection $(c, f(c))$, either $f''(c) = 0$ or $f''(c)$ fails to exist.

Example. Determine the concavity and find the inflection points of the function

$$f(x) = x^3 - 3x^2 + 2.$$

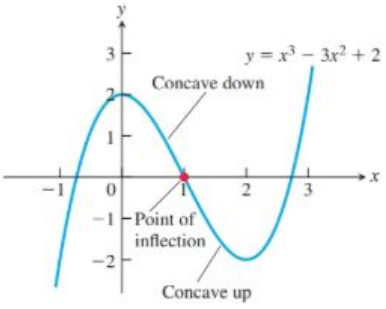


FIGURE 4.27 The concavity of the graph of f changes from concave down to concave up at the inflection point (Example 3).

Example. The graph of $f(x) = x^{\frac{5}{3}}$ has a horizontal tangent at the origin because

$$f'(x) = \frac{5}{3}x^{\frac{2}{3}} = 0 \Rightarrow x = 0.$$

However, the second derivative,

$$f''(x) = \frac{d}{dx}\left(\frac{5}{3}x^{\frac{2}{3}}\right) = \frac{10}{9}x^{-\frac{1}{3}},$$

Undefined at $x = 0$.

Note: $f''(x) < 0$ for $x < 0$ and $f''(x) > 0$ for $x > 0$, so the second derivative changes sign at $x = 0$, and there is a point of inflection at the origin.

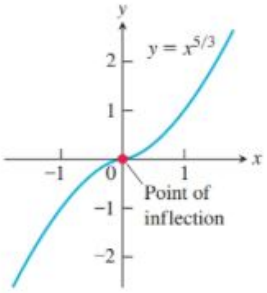
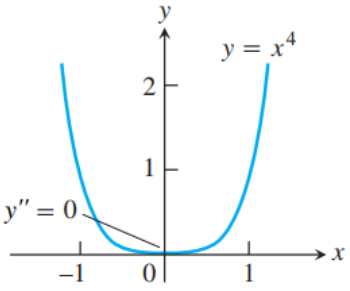


FIGURE 4.28 The graph of $f(x) = x^{5/3}$ has a horizontal tangent at the origin where the concavity changes, although f'' does not exist at $x = 0$ (Example 4).

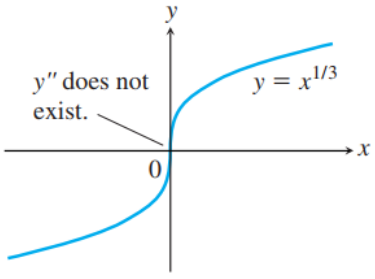
Example. *An Inflection Point May No Exist Where* $y'' = 0$.



The curve $y = x^4$ has no inflection point at $x = 0$.

Even though $y'' = 12x^2$ is zero there, it does no change sign.

Example. *An Inflection Point May Occur Where* y'' *Does Not Exist.*



The curve $y = x^{\frac{1}{3}}$ has no inflection at $x = 0$, but y'' does not exist there.

$$y'' = \frac{d^2}{dx^2}\left(x^{\frac{1}{3}}\right) = \frac{d}{dx}\left(\frac{1}{3}x^{-\frac{2}{3}}\right) = -\frac{2}{9}\left(x^{-\frac{5}{3}}\right).$$

We note that y'' can never be zero but y'' does not exists at $x = 0$.

Caution Example 4 in Section 4.1 (Figure 4.9) shows that the function $f(x) = x^{\frac{2}{3}}$ does not have a second derivative at $x = 0$ and does not have a point of inflection there(there is no change in concavity at $x = 0$.

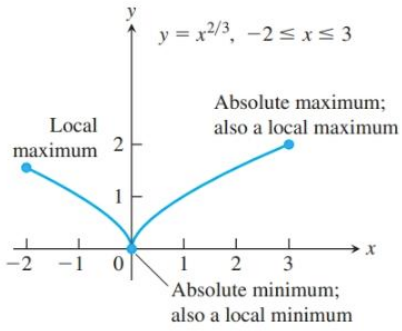


FIGURE 4.9 The extreme values of $f(x) = x^{2/3}$ on $[-2, 3]$ occur at $x = 0$ and $x = 3$ (Example 4).

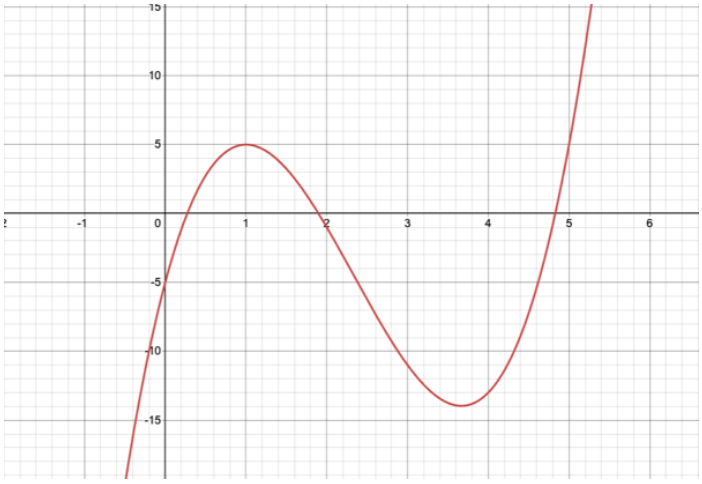
Combined with the behaviour of the function in Example above, we see that when the second derivative does not exist at $x = c$, an inflection point may or may not occur there.

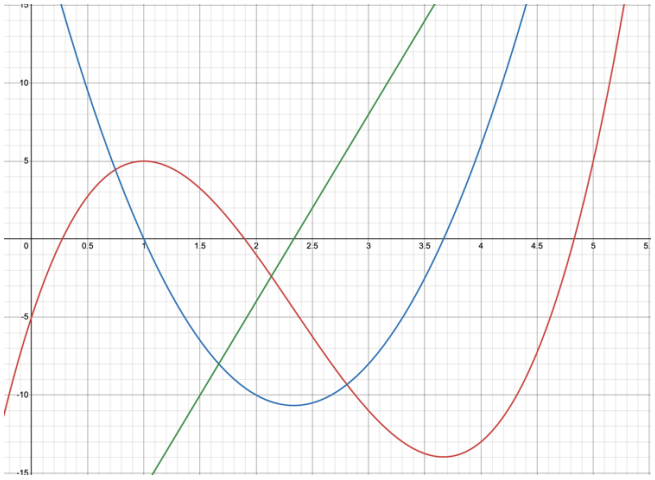
So we need to be careful about interpreting functional behaviour whenever first or second derivatives fail to exist at a point. At such points the graph can have vertical tangent lines, corners, cusps, or various discontinuities.

Example. A particle is moving along a horizontal coordinate line (positive to the right) with position function

$$s(t) = 2t^3 - 14t^2 + 22t - 5, \quad t \ge 0.$$

Find the velocity and acceleration, and describe the motion of particle.





Theorem. *Second Derivative Test for Local Extrema*

Suppose f'' is continuous on an open interval that contains $x = c$.
1. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

Note. This test requires us to know f'' *only at* c *itself* and not in an interval about c. This makes the test easy to apply. That's the good news. The bad news is that the test is inconclusive if $f'' = 0$ or if f'' does not exist at $x = c$. When this happens, use the First Derivative Test for local extreme values.

Graph Sketching

Example. *Using* f' *and* f'' *to Graph* f.

Sketch a graph of the function $f(x) = x^4 - 4x^3 + 10$ by using the guided steps.

(a) Identify where the extrema of f occur.
(b) Find the intervals on which f is increasing and the intervals on which f is decreasing.
(c) Find where the graph of f is concave up and where it is concave down.
(d) Sketch the general shape of the graph for f.
(e) Plot some specific points, such as local maximum and minimum points, points of inflection, and intercepts.
Then sketch the curve.

Solution.
We note that f is continuous on $(-\infty, \infty)$, because f is a polynomial. f is not even or odd function.
$f'(x) = 4x^3 - 12x^2$, f' is continuous on $(-\infty, \infty)$,
$f''(x) = 12x^2 - 24x$, f'' is also continuous on $(-\infty, \infty)$.

For critical points, $f'(x) = 4x^2(x-3) = 0$, that is $x = 0$ $(y = 10)$ and $x = 3$ $(y = -17)$.

Intervals	$x < 0$	$0 < x < 3$	$x > 3$
Sign of f'	$-$	$-$	$+$
Behaviour of f	Decreasing	Decreasing	Increasing

Using the First Derivative Test for local extrema and the table above, we see that there is no extremum at $x = 0$ and a local minimum at $x = 3$. We see that f is decreasing on $(-\infty, 0]$ and $[0,3]$ and increasing on $[3, \infty)$.

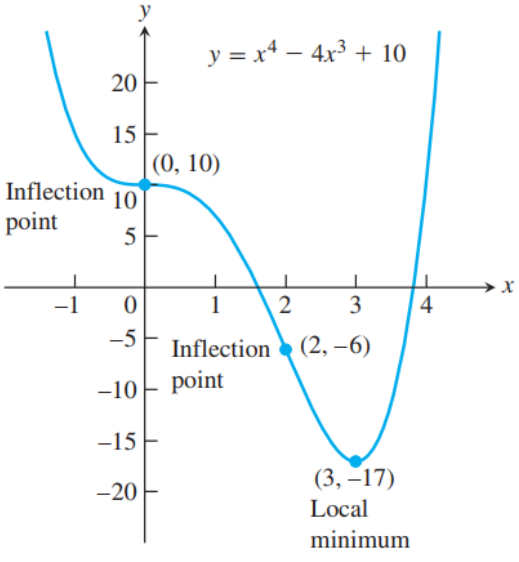
$f''(x) = 12x(x-2) = 0$, that is $x = 0$ $(y = 10)$ and $x = 2$ $(y = -6)$.

Intervals	$x < 0$	$0 < x < 2$	$x > 2$
Sign of f''	$+$	$-$	$+$
Behaviour of f	Concave up	Concave down	Concave up

We see that f is concave up on the intervals $(-\infty, 0)$ and $(2,0)$, and concave down on $(0,2)$.

$x < 0$	$0 < x < 2$	$2 < x < 3$	$x > 3$
Decreasing	Decreasing	Decreasing	Increasing
Concave up	Concave down	Concave up	Concave up

There are no asymptotes in this graph and when $x = 0, y = 10$. We are ready to plot the graph.



Procedure for Graphing $y = f(x)$.

1. Identify the domain of f and any symmetries the curve may have.
2. Find y' and y''.
3. Find the critical points of f, and identify the function's behaviour at each of them.
4. Find where the curve is increasing and where it is decreasing.
5. Find the points of inflection, if any occur, and determine the concavity of the curve.
6. Identify any asymptotes.
7. Plot key points, such as the interceptions and the points found in Steps 3-5, and sketch the curve.

Example. *Using the Graphing Strategy*
Sketch the graph of

$$f(x) = \frac{(x+1)^2}{1+x^2}.$$

Solution. The domain of f is $(-\infty, \infty)$ and there are no symmetries about either axis or the origin (not even or odd function).

Find f' *and* f''.

$$f'(x) = \frac{(1+x^2)\big(2(x+1)\big) - (x+1)^2(2x)}{(1+x^2)^2} = \frac{2(1-x^2)}{(1+x^2)^2}.$$

$$f''(x) = \frac{(1+x^2)^2\big(2(-2x)\big) - 2(1-x^2)\big(2(1+x^2)(2x)\big)}{(1+x^2)^4} = \frac{4x(x^2-3)}{(1+x^2)^3}.$$

Behaviour at Critical Points
For $f'(x) = 0 \Rightarrow\ x = \pm 1$ and f' is defined everywhere over the domain f. By Second Derivative Test, $f''(1) = \frac{4(-2)}{2^3} = -1 < 0$ (maximum) and $f''(-1) = \frac{-4(-2)}{2^3} = 1 > 0$ (minimum).
Increasing and decreasing

$$f'(x) = \frac{2(1-x^2)}{(1+x^2)^2} = \frac{2(1-x)(1+x)}{(1+x^2)^2}.$$

Intervals	$x < -1$	$-1 < x < 1$	$x > 1$
Sign of f'	$-$	$+$	$-$
Behaviour of f	Decreasing	Increasing	Decreasing

We obtain, $(-1,0)$ is a local minimum point, $(1,2)$ is a local maximum point.

Inflection points
For $f''(x) = 0 \Rightarrow x = 0, \pm\sqrt{3}$ and denominator of f'' always positive and never be zero.

$$f''(x) = \frac{4x(x^2-3)}{(1+x^2)^3} = \frac{4x(x-\sqrt{3})(x+\sqrt{3})}{(1+x^2)^3}$$

Intervals	$x < -\sqrt{3}$	$-\sqrt{3} < x < 0$	$0 < x < \sqrt{3}$	$x > \sqrt{3}$
Sign of f''	$-$	$+$	$-$	$+$
Behaviour of f	Concave Downward	Concave Upward	Concave Downward	Concave Upward

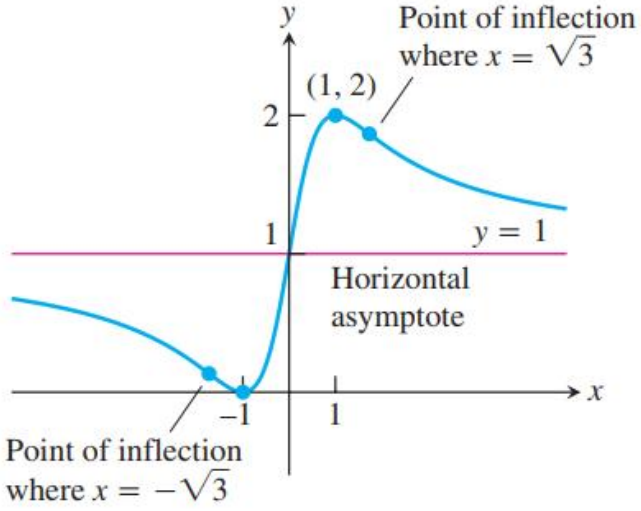
We obtain, $\left(-\sqrt{3}, \frac{1-\sqrt{3}}{4}\right)$, $(0,1)$ and $\left(\sqrt{3}, \frac{1+\sqrt{3}}{4}\right)$ are inflection points.

Asymptotes
We have

$$f(x) = \frac{(x+1)^2}{1+x^2} = \frac{x^2+2x+1}{1+x^2} = 1 + \frac{2x}{1+x^2}$$

As $x \to \infty, f(x) \to 1^+$ and $x \to -\infty, f(x) \to 1^-$. Hence, $y = 1$ is an asymptote.

We have $(0,1)$ is a interception point. We are ready to sketch the graph.



Example
Sketch the graph of

$$f(x) = \frac{x^2 + 4}{2x}.$$

Extra Exercises:
Sketch the graphs for
(a) $f(x) = \frac{x^2}{4-x^2}$,

(b) $f(x) = x\sqrt{6 - x}$

Antiderivatives.

We have studied how to find the derivative of a function. However, many problems require that we recover a function from its known derivative (from its known rate of change).
For instance, the laws of physics tell us the acceleration of an object falling from an initial height, and we can use this to compute its velocity and its height at any time.
Generally, we want to find a function F from its derivative f. If such a function F exists, it is called an antiderivative of f.
Antiderivatives have an important connection to the theory of integrals that is developed in Chapter 5. For this reason the process of taking an antiderivative is also called "integration."

Definition. *Antiderivative*

A function F is an ***antiderivative*** of f on an interval I if $F'(x) = f(x)$ for all x in I.

The process of recovering a function $F(x)$ from its derivative $f(x)$ is called *antidifferentiation*. We use capital letters such as F to represent an antiderivative of a function f, G to represent an antiderivative of g, and so forth.

Example. *Finding Antiderivatives*
Find an antiderivative for each of the following functions.

(a) $f(x) = 2x$
(b) $g(x) = \cos x$
(c) $h(x) = \sec^2 x + \frac{1}{2\sqrt{x}}$

Solution
(a) $F(x) = x^2$
(b) $G(x) = \sin x$
(c) $H(x) = \tan x + \sqrt{x}$.

Each answer can be checked by differentiating. The derivative of $F(x) = x^2$ is $2x$ and so on.
The function $x^2 + 1$ has the same derivative. So does $x^2 + C$ for any constant C. Are there others?
More generally, we have the following result.

If F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

Thus, the most general antiderivative of f on I is a *family* of functions $F(x) + C$ whose graphs are vertical translates of one another. We can select a particular antiderivative from this family by assigning a specific value to C.

Example. Find an antiderivative of $f(x) = 3x^2$ that satisfied $F(1) = -1$.

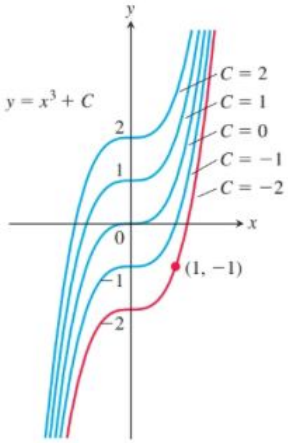


FIGURE 4.52 The curves $y = x^3 + C$ fill the coordinate plane without overlapping. In Example 2, we identify the curve $y = x^3 - 2$ as the one that passes through the given point $(1, -1)$.

Example. *Finding a Particular Antiderivative*
Find an antiderivative of $f(x) = \sin x$ that satisfied $F(0) = 3$.

Solution. Since the derivative of $-\cos x$ is $\sin x$, the general antiderivative

$$F(x) = -\cos x + C$$

Gives all the antiderivatives of $f(x)$. The condition $F(0) = 3$ determines a specific value for C.

We substituting $x = 0$ into $F(x) = -\cos 0 + C$ gives

$$F(0) = -\cos 0 + C = -1 + C.$$

Since, $F(0) = 3$. It gives $C = 4$. So,

$$F(x) = -\cos x + 4$$

is the antiderivative satisfying $F(0) = 3$.

The following gives antiderivative formulas for a number of important functions.

TABLE 4.2 Antiderivative formulas, k a nonzero constant

Function	General antiderivative
1. x^n	$\frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$
2. $\sin kx$	$-\frac{1}{k}\cos kx + C$
3. $\cos kx$	$\frac{1}{k}\sin kx + C$
4. $\sec^2 kx$	$\frac{1}{k}\tan kx + C$
5. $\csc^2 kx$	$-\frac{1}{k}\cot kx + C$
6. $\sec kx \tan kx$	$\frac{1}{k}\sec kx + C$
7. $\csc kx \cot kx$	$-\frac{1}{k}\csc kx + C$

Example: Find the general antiderivative of each of the following functions.

(a) $f(x) = x^5$

(b) $g(x) = \frac{1}{\sqrt{x}}$

(c) $h(x) = \sin 2x$

(d) $f(x) = \cos\left(\frac{x}{2}\right)$

Other derivative rules also lead to corresponding antiderivative rules. We can add and subtract antiderivatives and multiply them by constants.

TABLE 4.3 Antiderivative linearity rules

	Function	General antiderivative
1. *Constant Multiple Rule*:	$kf(x)$	$kF(x) + C$, k a constant
2. *Sum or Difference Rule*:	$f(x) \pm g(x)$	$F(x) \pm G(x) + C$

The formulas in Table 4.3 are easily proved by differentiating the antiderivatives and verifying that the result agrees with the original function.

Example: Find the general antiderivative of

$$f(x) = \frac{3}{\sqrt{x}} + \sin 2x.$$

Initial Value Problems and Differential Equations

Antiderivatives play several important roles in mathematics and its applications. Methods and techniques for finding them are a major part of calculus. Finding an antiderivative for a function $f(x)$ is the same problem as finding a function $y(x)$ that satisfies the equation

$$\frac{dy}{dx} = f(x).$$

This is called a ***differential equation***, since it is an equation involving an unknown function y that is being differentiated.

To solve it, we need a function $y(x)$ that satisfies the equation. This function is found by taking the antiderivative of $f(x)$. We can fix the arbitrary constant arising in the antidifferentiation process by specifying an initial condition

$$y(x_0) = y_0.$$

This condition means the function $y(x)$ has the value y_0 when $x = x_0$. The combination of a differential equation and an initial condition is called an **initial value problem**.

Antiderivatives and Motion

Example: A hot-air balloon ascending at the rate of 3.6 m/s is at a height 24.5 m above the ground when a package is dropped. How long does it take the package to reach the ground?

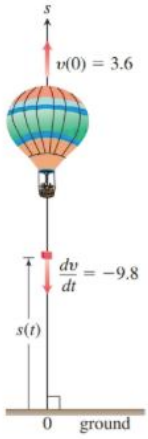


FIGURE 4.53 A package dropped from a rising hot-air balloon (Example 5).

Indefinite Integrals

A special symbol is used to denote the collection of all antiderivatives of a function f.

Definition. The collection of all antiderivatives of f is called the **indefinite integral** of f with respect to x; it is denoted by

$$\int f(x)\,dx.$$

The symbol $\int$ is an **integral sign**. The function f is the **integrand** of the integral, and x is the **variable of integration**.

After the integral sign in the notation we just defined, the integrand function is always followed by a differential to indicate the variable of integration.

Using this notation, we restate the solutions of Example.

$$\int 2x\,dx = x^2 + C,$$

$$\int \cos x\,dx = \sin x + C,$$

$$\int \left(\sec^2 x + \frac{1}{2\sqrt{x}}\right) dx = \tan x + \sqrt{x} + C.$$

Example: Evaluate

$$\int (x^2 - 2x + 5)\,dx.$$

TABLE 4.4 Integration formulas

1. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
2. $\int \sin x\,dx = -\cos x + C$
3. $\int \cos x\,dx = \sin x + C$
4. $\int \sec^2 x\,dx = \tan x + C$
5. $\int \csc^2 x\,dx = -\cot x + C$
6. $\int \sec x \tan x\,dx = \sec x + C$
7. $\int \csc x \cot x\,dx = -\csc x + C$